



COS20028 – BIG DATA ARCHITECTURE AND APPLICATION

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Week 5



Writing MapReduce Program -III



DEVELOPMENT TIPS AND TECHNIQUES

Strategies for Debugging MapReduce Code



Debugging MapReduce code is not easy.

Each instance of a Mapper runs as a separate task (Often on a different machine).

Difficult to attach a debugger to the process.

Difficult to catch “edge cases.”

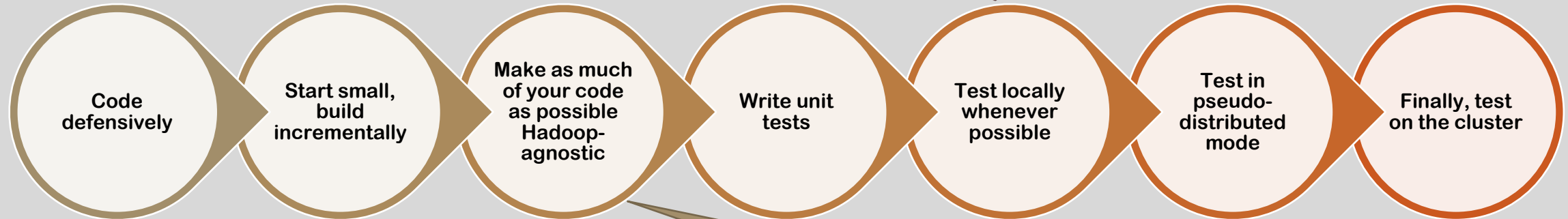


Very large volumes of data means that unexpected input is likely to appear

Code which expects all data to be well-formed is likely to fail.

Debugging Tips

- With sampled small-scale data.



- Ensure that input data is in the expected format.
- Expect things to go wrong and catch exceptions.

- Make it easier to test.

Testing Strategies

When testing in pseudo-distributed mode, ensure that you are testing with a similar environment to that on the real cluster.

- **Same amount of RAM allocated to the task JVMs.**
- **Same Hadoop version.**
- **Same Java version.**
- **Same third-party library versions.**

Testing Locally Using LocalJobRunner

Hadoop can run Mapreduce in a single, local process.

- Does not require any Hadoop daemons to be running.
- Uses the local file system instead of HDFS.
- Known as LocalJobRunner mode.

This is a very useful way for quickly testing incremental changes to code.

Testing via LocalJobRunner

- To run in LocalJobRunner mode, add the following lines to the driver code:

```
Configuration conf = new Configuration();  
conf.set("mapred.job.tracker", "local");  
conf.set("fs.default.name", "file:///");
```

← In Hadoop 1.x only.

```
Configuration conf = new Configuration();  
conf.set("mapreduce.framework.name", "local");  
conf.set("fs.defaultFS", "file:///");  
Job job = Job.getInstance(getConf(), "Code Local Job Runner Word Count");
```

← In Hadoop 2.x and above.

- Or, set these options on the command line if your driver uses ToolRunner

- -fs is equivalent to -D *fs.default.name*
- -jt is equivalent to -D *mapred.job.tracker*

← In Hadoop 1.x only.

hadoop jar *yourjar.jar YourDriver* -fs=file:/// -jt=local *Input Output*

```
hadoop jar ljrwc.jar stubs.LocalWordCount -fs=file:/// -jt=local shakespeare.tar.gz cmljrwc_res
```


Limitations of LocalJobRunner Mode

- Distributed Cache does NOT work.
- The job can only specify a single Reducer.
- Some “beginner” mistakes may not be caught. For example, attempting to share data between Mappers will work (which it should not be valid in the real situation) because the code is running in a single JVM.

LocalJobRunner Mode in Eclipse

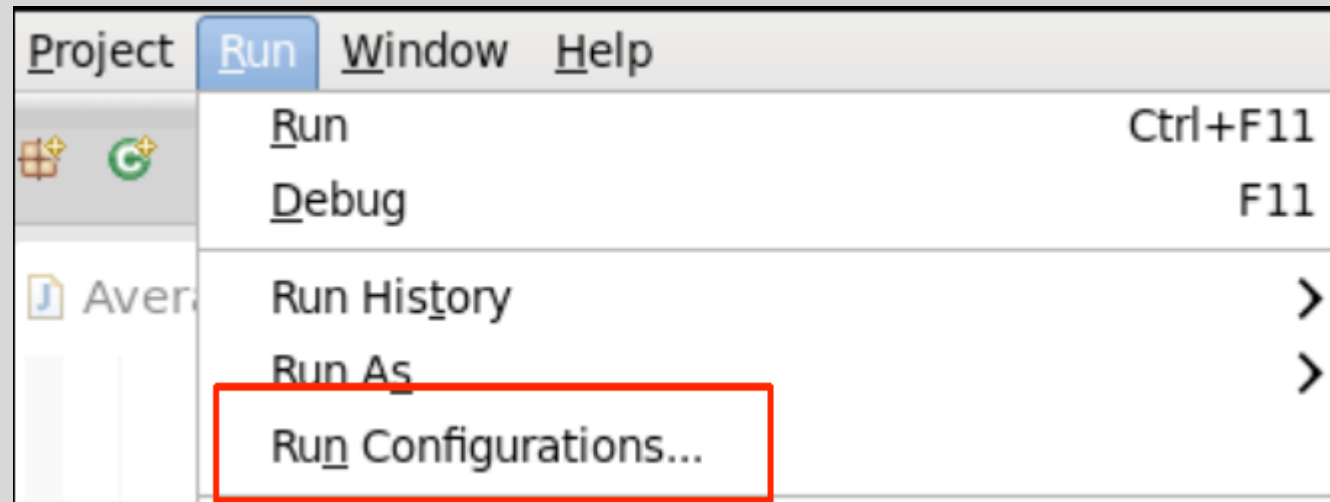
In the VM we used in the lab, the Eclipse can also run Hadoop code in LocalJobRunner mode from within the IDE.

- This is Hadoop's default behaviour when no configuration is provided.

This allows rapid development iterations. (Agile programming)

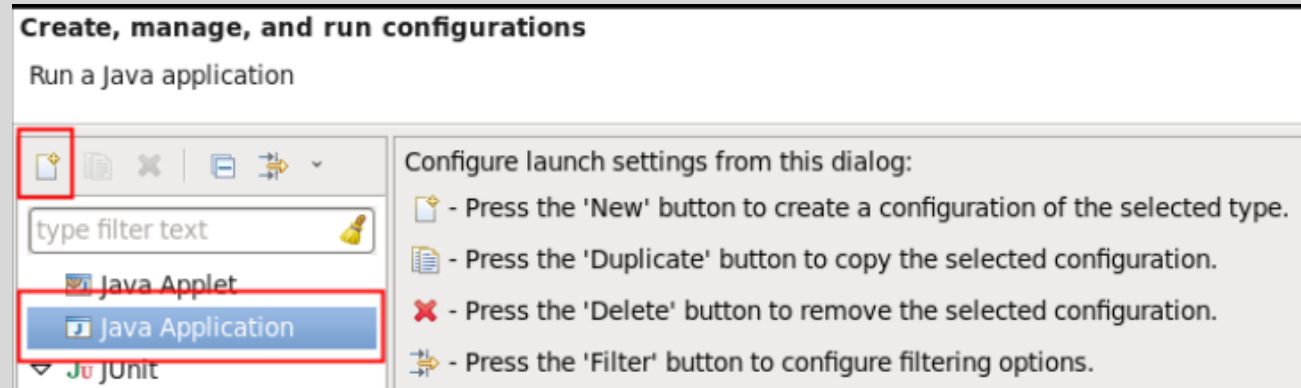
LocalJobRunner Mode in Eclipse

- Find the corresponding setting in Eclipse

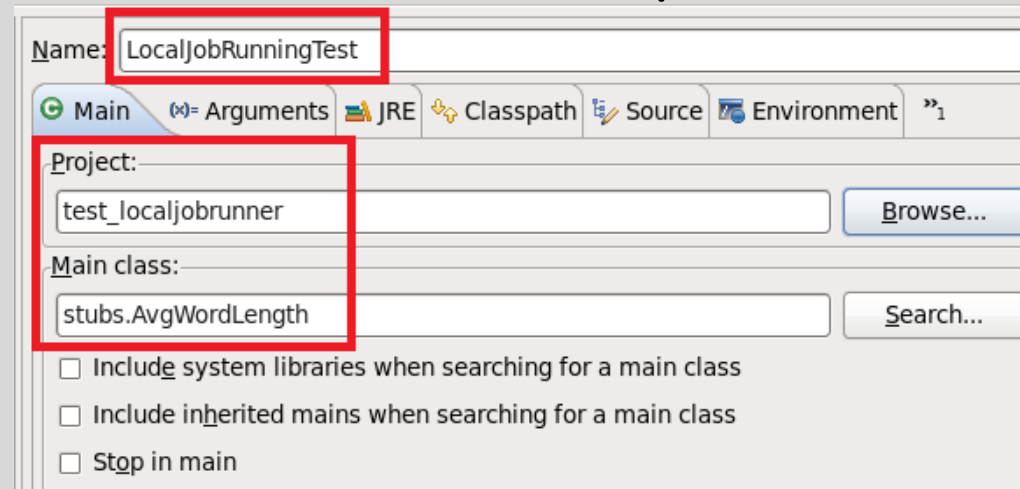


LocalJobRunner Mode in Eclipse

- Select Java Application and then click on the “New” button.

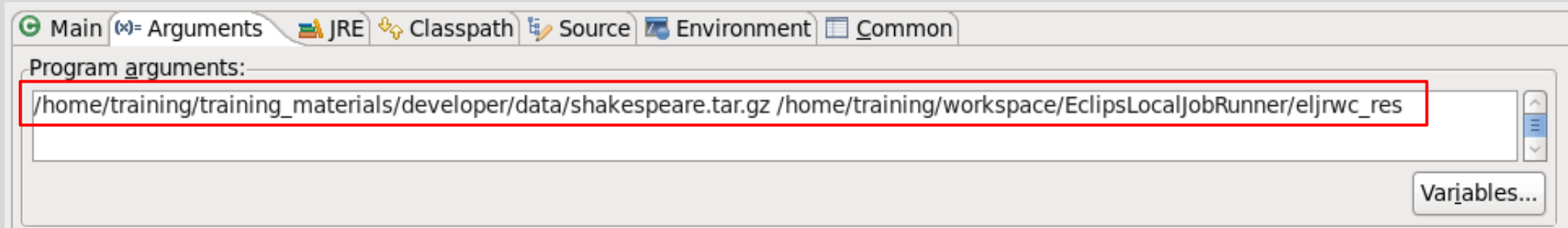


- Verify the Project and Main Class fields are pre-filled correctly.

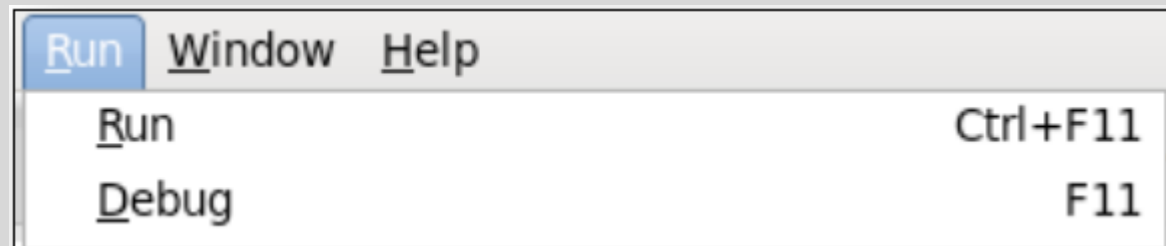


LocalJobRunner Mode in Eclipse

- Specify values in the Arguments tab.
 - Local input/output file
 - Any configuration options needed when your job runs.

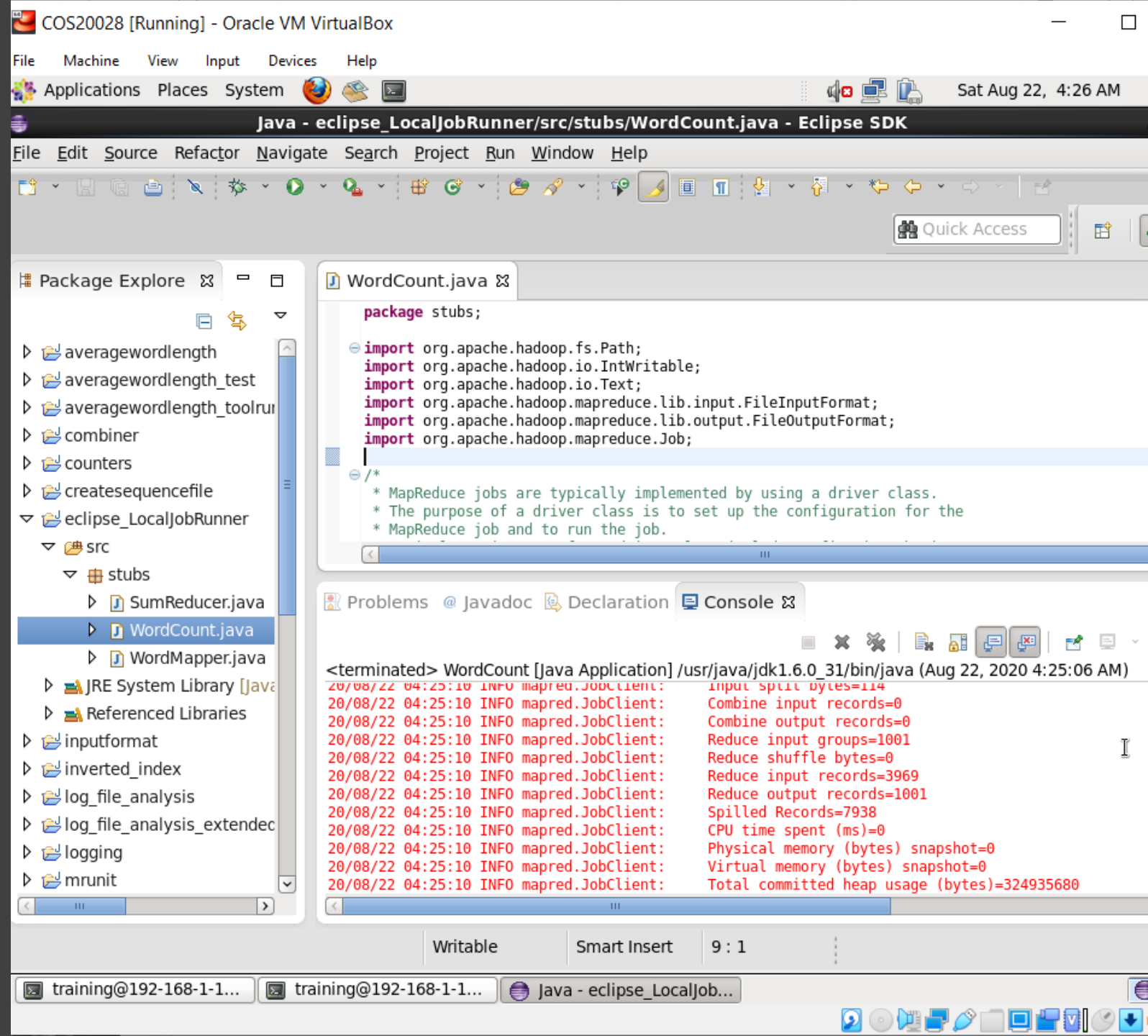


- Define breakpoints if desired.
- Execute the application in run mode or debug mode.



LOCALJOBRUNNER MODE IN ECLIPSE

Review output in the Eclipse
console window





WRITING AND VIEWING LOG FILES

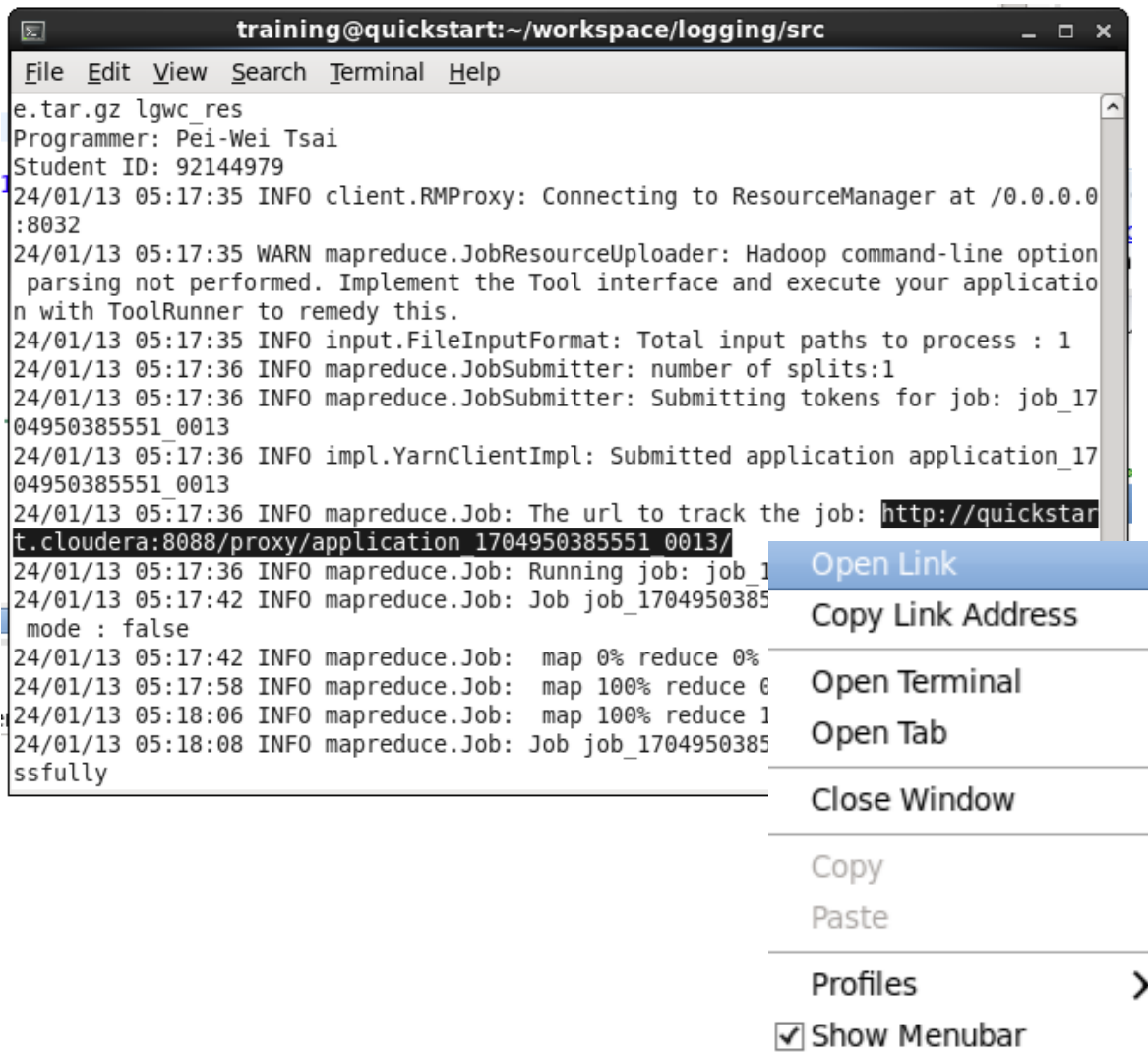
Using *stdout* and *stderr*

Tried-and-true debugging technique: write to *stdout* or *stderr*

- If running in LocalJobRunner mode, you will see the results of *System.err.println()*

If running on a cluster, that output will not appear on your console

- Output is visible via Hadoop's Web UI.



The screenshot shows a terminal window titled "training@quickstart:~/workspace/logging/src". The terminal displays logs for a Hadoop Yarn job submission. A right-click context menu is open over the URL "http://quickstart.t.cloudera:8088/proxy/application_1704950385551_0013/". The menu options are: Open Link, Copy Link Address, Open Terminal, Open Tab, Close Window, Copy, Paste, Profiles, and Show Menubar (checked).

```
File Edit View Search Terminal Help
e.tar.gz lgwc_res
Programmer: Pei-Wei Tsai
Student ID: 92144979
24/01/13 05:17:35 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/01/13 05:17:35 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
24/01/13 05:17:35 INFO input.FileInputFormat: Total input paths to process : 1
24/01/13 05:17:36 INFO mapreduce.JobSubmitter: number of splits:1
24/01/13 05:17:36 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1704950385551_0013
24/01/13 05:17:36 INFO impl.YarnClientImpl: Submitted application application_1704950385551_0013
24/01/13 05:17:36 INFO mapreduce.Job: The url to track the job: http://quickstart.t.cloudera:8088/proxy/application_1704950385551_0013/
24/01/13 05:17:36 INFO mapreduce.Job: Running job: job_1704950385551_0013
24/01/13 05:17:42 INFO mapreduce.Job: Job job_1704950385551_0013 mode : false
24/01/13 05:17:42 INFO mapreduce.Job: map 0% reduce 0%
24/01/13 05:17:58 INFO mapreduce.Job: map 100% reduce 0%
24/01/13 05:18:06 INFO mapreduce.Job: map 100% reduce 100%
24/01/13 05:18:08 INFO mapreduce.Job: Job job_1704950385551_0013 completed successfully
```

Hadoop Yarn Job Tracking

- Hadoop Yarn in an integrated platform, which provides the job tracking function.
- When executing your job, you will see a URL pops-up in the displayed information.
- Right-click the URL and click “Open Link” to see the tracking result.

LOGFILE VIA WEB UI

5.13_with_code [Running] - Oracle VM VirtualBox

Sat Jan 13, 5:23 AM training

MapReduce Job job_1704950385551_0013 - Mozilla Firefox

MapReduce Job job_170... x

quickstart.cloudera:19888/jobhistory/job/job_1704950385551_0013

Hue HDFS

hadoop

MapReduce Job job_1704950385551_0013

Logged in as: dr.who

Application

Job

Overview

Counters

Configuration

Map tasks

Reduce tasks

Tools

Job Overview

Job Name: Logging Word Count

User Name: training

Queue: root.training

State: SUCCEEDED

Uberized: false

Submitted: Sat Jan 13 05:17:36 PST 2024

Started: Sat Jan 13 05:17:41 PST 2024

Finished: Sat Jan 13 05:18:05 PST 2024

Elapsed: 24sec

Diagnostics:

Average Map Time: 13sec

Average Shuffle Time: 3sec

Average Merge Time: 0sec

Average Reduce Time: 1sec

ApplicationMaster			
Attempt Number	Start Time	Node	Logs
1	Sat Jan 13 05:17:37 PST 2024	quickstart.cloudera:8042	logs

Task Type	Total	Complete
Map	1	1
Reduce	1	1

Attempt Type	Failed	Killed	Successful
Maps	0	0	1
Reduces	0	0	1

training@quickstart:~... | java - logging/src/stub... | MapReduce Job job_17...

Right Ctrl

LOGFILE VIA WEB UI

5.13_with_code [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Sat Jan 13, 5:26 AM training Applications Places System

SUCCESSFUL MAP attempts in job_1704950385551_0013 - Mozilla Firefox

quickstart.cloudera:19888/jobhistory/attempts/job_1704950385551_0013/m/SUCCESSFUL

Hue HDFS

hadoop

SUCCESSFUL MAP attempts in job_1704950385551_0013

Logged in as: drwho

Application

Job

Tools

Configuration

Local logs

Server stacks

Server metrics

Show: 20 entries

Search:

Attempt	State	Status	Node	Logs	Start Time	Finish Time	Elapsed Time	Note
attempt_1704950385551_0013_m_000000_0	SUCCEEDED	map	/default-rack/quickstart.cloudera:8042	logs	Sat Jan 13 05:17:43 -0800 2024	Sat Jan 13 05:17:57 -0800 2024	13sec	

Showing 1 to 1 of 1 entries

First Previous 1 Next Last

NodeManager information

Total Vmem allocated for Containers 16.80 GB

Vmem enforcement enabled false

Total Pmem allocated for Container 8 GB

Pmem enforcement enabled true

Total VCores allocated for Containers 8

NodeHealthyStatus true

LastNodeHealthTime Sat Jan 13 05:25:33 PST 2024

NodeHealthReport

Node Manager Version: 2.6.0-cdh5.13.0 from 42e8860b182e55321bd5f5605264da4dc8882be by jenkins source checksum 90d315b02cac6c524a5f659051adb221 on 2017-10-04T18:16Z

Hadoop Version: 2.6.0-cdh5.13.0 from 42e8860b182e55321bd5f5605264da4dc8882be by jenkins source checksum 5e84c185f8a22158e2b0e4b8f85311 on 2017-10-04T18:08Z

Log Type: syslog

Log Upload Time: Sat Jan 13 05:18:15 -0800 2024

Log Length: 31955681

2024-01-13 05:17:48,266 WARN [main] org.apache.hadoop.metrics2.impl.MetricsConfig: Cannot locate configuration: tried hadoop-metrics2-maptask.properties,hadoop-metrics2.properties

2024-01-13 05:17:48,649 INFO [main] org.apache.hadoop.metrics2.impl.MetricsSystemImpl: Scheduled snapshot period at 10 second(s).

2024-01-13 05:17:48,653 INFO [main] org.apache.hadoop.metrics2.impl.MetricsSystemImpl: MapTask metrics system started

2024-01-13 05:17:48,709 INFO [main] org.apache.hadoop.mapred.YarnChild: Executing with tokens:

2024-01-13 05:17:48,710 INFO [main] org.apache.hadoop.mapred.YarnChild: Kind: mapreduce.job, Service: job_1704950385551_0013, Ident: (org.apache.hadoop.mapreduce.security.token.JobTokenIdentifier@1e1a0486)

2024-01-13 05:17:49,582 INFO [main] org.apache.hadoop.mapred.YarnChild: Sleeping for 0ms before retrying again. Got null now.

2024-01-13 05:17:50,123 INFO [main] org.apache.hadoop.mapred.YarnChild: mapreduce.cluster.local.dir for child: /var/lib/hadoop-yarn/cache/yarn/nm-local-dir/usercache/training/appcache/application_1704950385551_0013

2024-01-13 05:17:50,549 INFO [main] org.apache.hadoop.conf.Configuration.deprecation: session.id is deprecated. Instead, use dfs.metrics.session-id

2024-01-13 05:17:52,196 INFO [main] org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter: File Output Committer Algorithm version is 1

2024-01-13 05:17:52,199 INFO [main] org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false

2024-01-13 05:17:52,301 INFO [main] org.apache.hadoop.mapred.Task: Using ResourceCalculatorProcessTree : []

2024-01-13 05:17:52,699 INFO [main] org.apache.hadoop.mapred.MapTask: Processing split: hdfs://quickstart.cloudera:8020/user/training/shakespeare.tar.gz:0+2080857

2024-01-13 05:17:53,165 INFO [main] org.apache.hadoop.mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)

2024-01-13 05:17:53,165 INFO [main] org.apache.hadoop.mapred.MapTask: mapreduce.task.io.sort.mb: 100

2024-01-13 05:17:53,165 INFO [main] org.apache.hadoop.mapred.MapTask: soft limit at 83886080

2024-01-13 05:17:53,165 INFO [main] org.apache.hadoop.mapred.MapTask: bufstart = 0; bufvoid = 104857600

2024-01-13 05:17:53,167 INFO [main] org.apache.hadoop.mapred.MapTask: kvstart = 26214396; length = 6553600

2024-01-13 05:17:53,188 INFO [main] org.apache.hadoop.mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask\$MapOutputBuffer

2024-01-13 05:17:53,236 INFO [main] org.apache.hadoop.io.compress.zlib.ZlibFactory: Successfully loaded & initialized native-zlib library

2024-01-13 05:17:53,237 INFO [main] org.apache.hadoop.io.compress.zlib.ZlibFactory: Got brand-new decompressor [.gz]

2024-01-13 05:17:53,270 INFO [main] stubs.WordMapper: info1

2024-01-13 05:17:53,270 WARN [main] stubs.WordMapper: warn1

2024-01-13 05:17:53,270 ERROR [main] stubs.WordMapper: error1

2024-01-13 05:17:53,271 INFO [main] stubs.WordMapper: info1

2024-01-13 05:17:53,271 WARN [main] stubs.WordMapper: warn1

Why Logging is better than Printing?

println statements rapidly become awkward

- Turning them on and off in your code is tedious, and leads to errors.

Logging provides much finer-grained control over:

- What gets logged.
- When something gets logged.
- How something is logged.



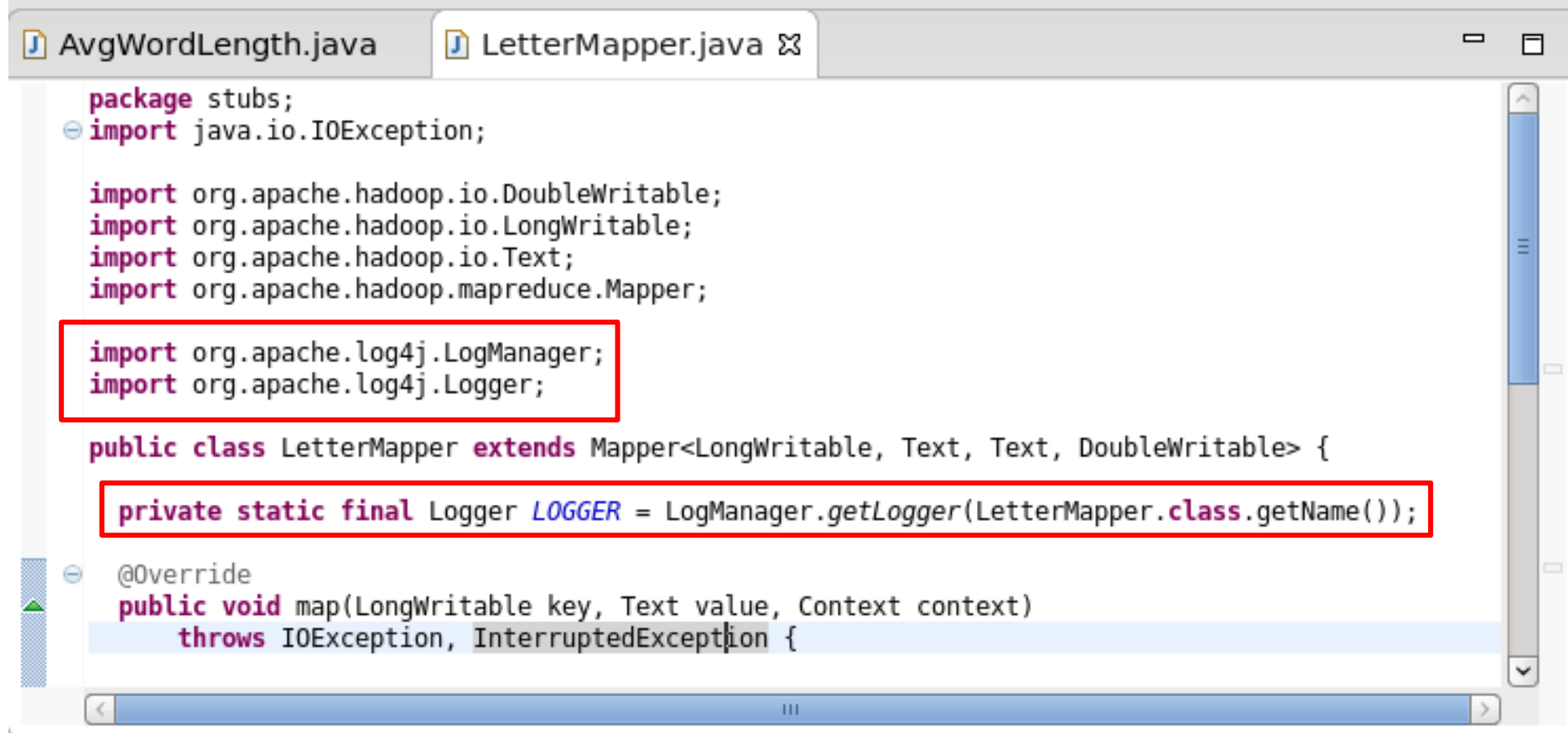
log4j Library

- Hadoop uses *log4j* to generate all its log files.
- Your Mappers and Reducers can also use *log4j*.
 - All the initialisation is handled for you by Hadoop.
- Add the *log4j.jar-**<version>*** file from your CDH distribution to your classpath when you reference the *log4j* classes.

log4j Library

Simply send strings to loggers tagged with severity levels

- **LOGGER.trace(*“message”*);**
- **LOGGER.debug(*“message”*);**
- **LOGGER.info(*“message”*);**
- **LOGGER.warn(*“message”*);**
- **LOGGER.error(*“message”*);**



```
package stubs;
import java.io.IOException;

import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

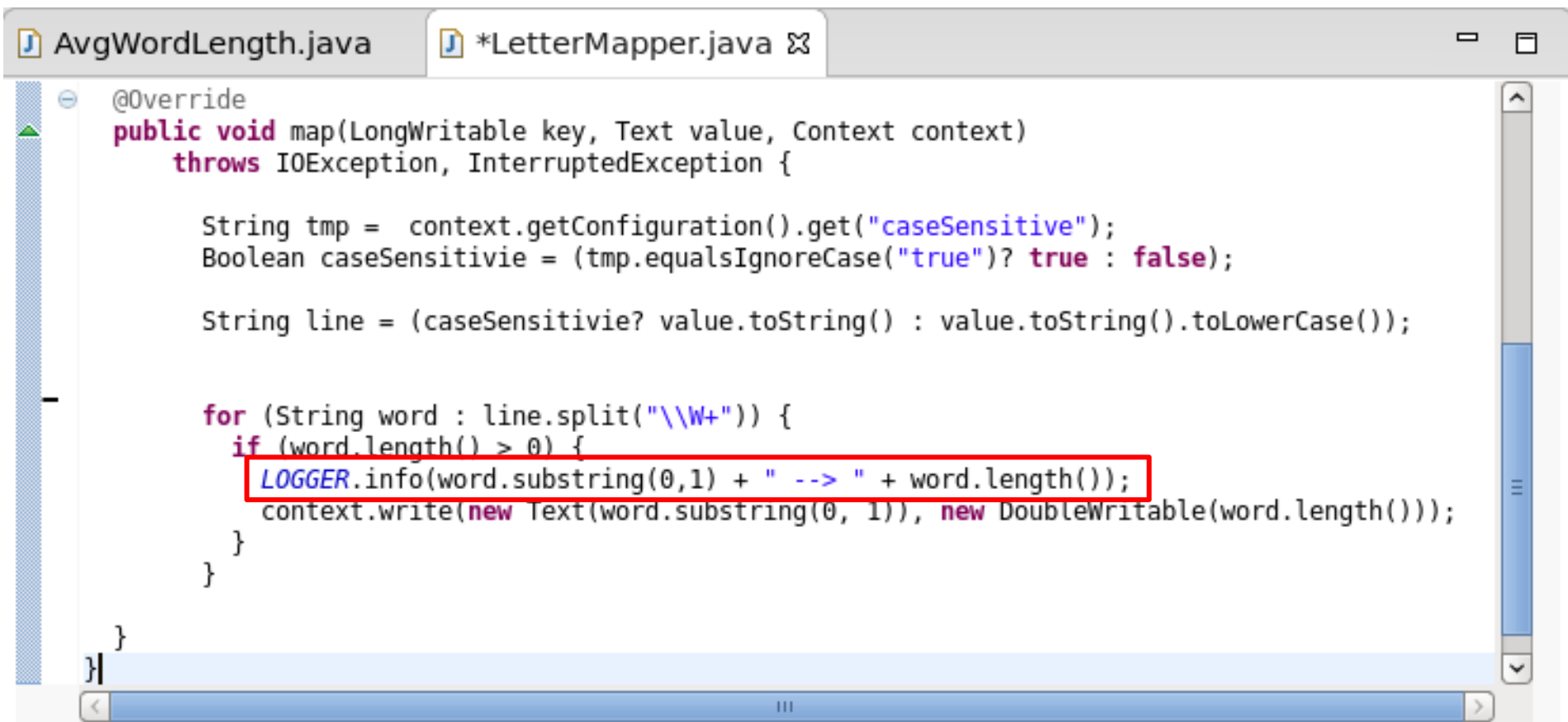
import org.apache.log4j.LogManager;
import org.apache.log4j.Logger;

public class LetterMapper extends Mapper<LongWritable, Text, Text, DoubleWritable> {

    private static final Logger LOGGER = LogManager.getLogger(LetterMapper.class.getName());

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
```

USING *LOG4J* TO STORE LOGS



```
@Override
public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {

    String tmp = context.getConfiguration().get("caseSensitive");
    Boolean caseSensitive = (tmp.equalsIgnoreCase("true")? true : false);

    String line = (caseSensitive? value.toString() : value.toString().toLowerCase());

    for (String word : line.split("\\W+")) {
        if (word.length() > 0) {
            LOGGER.info(word.substring(0,1) + " --> " + word.length());
            context.write(new Text(word.substring(0, 1)), new DoubleWritable(word.length()));
        }
    }
}
```

USING *LOG4J* TO STORE LOGS

USING *LOG4J* TO STORE LOGS



▼ Application

[About](#)
[Jobs](#)

► Tools

Log Type: stderr

Log Upload Time: Sat Jan 13 05:49:00 -0800 2024

Log Length: 0

Log Type: stdout

Log Upload Time: Sat Jan 13 05:49:00 -0800 2024

Log Length: 0

Log Type: syslog

Log Upload Time: Sat Jan 13 05:49:00 -0800 2024

Log Length: 62366633

Showing 4096 bytes of 62366633 total. Click [here](#) for the full log.

tterMapper: t --> 3

```
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: m --> 6
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: h --> 5
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: m --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: w --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: s --> 5
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: h --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: b --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: m --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: I --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: s --> 6
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: b --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: n --> 3
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: b --> 3
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: t --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: m --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: f --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: i --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: t --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: S --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: g --> 6
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: g --> 5
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: o --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: W --> 10
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: w --> 5
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: h --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: T --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: t --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: I --> 1
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: l --> 2
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: s --> 5
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: a --> 3
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: s --> 4
2024-01-13 05:48:40,095 INFO [main] stubs.LetterMapper: a --> 5
```

```

1 package stubs;
2 import java.io.IOException;
3
4 import org.apache.hadoop.io.LongWritable;
5 import org.apache.hadoop.io.IntWritable;
6 import org.apache.hadoop.io.Text;
7 import org.apache.hadoop.mapreduce.Mapper;
8 import org.apache.log4j.Level;
9 import org.apache.log4j.LogManager;
10 import org.apache.log4j.Logger;
11
12 public class LetterMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
13
14     private static final Logger LOGGER = LogManager.getLogger(LetterMapper.class.getName());
15
16     @Override
17     public void map(LongWritable key, Text value, Context context)
18         throws IOException, InterruptedException {
19
20         LOGGER.setLevel(Level.ALL);
21         //LOGGER.setLevel(Level.DEBUG);
22         //LOGGER.setLevel(Level.ERROR);
23         //LOGGER.setLevel(Level.FATAL);
24         //LOGGER.setLevel(Level.INFO);
25         //LOGGER.setLevel(Level.TRACE);
26         //LOGGER.setLevel(Level.WARN);
27
28         String line = value.toString();
29
30         for (String word : line.split("\\W+")) {
31             if (word.length() > 0) {
32                 LOGGER.debug(word.substring(0,1) + " --> " + word.length());
33                 context.write(new Text(word.substring(0,1)), new IntWritable(word.length()));
34             }
35         }
36     }
37 }
38 }

```

Avoiding Expensive Operations

- By sending the parameter via the ToolRunner, we can decide whether display the debugging information without modifying the code.

```

1 package stubs;
2 import java.io.IOException;
3 import org.apache.hadoop.io.DoubleWritable;
4 import org.apache.hadoop.io.IntWritable;
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7 import org.apache.log4j.Level;
8 import org.apache.log4j.LogManager;
9 import org.apache.log4j.Logger;
10
11 public class AverageReducer extends Reducer<Text, IntWritable, Text, DoubleWritable> {
12
13     private static final Logger LOGGER = LogManager.getLogger(AverageReducer.class.getName());
14
15     @Override
16     public void reduce(Text key, Iterable<IntWritable> values, Context context)
17         throws IOException, InterruptedException {
18
19         LOGGER.setLevel(Level.ALL);
20         //LOGGER.setLevel(Level.DEBUG);
21         //LOGGER.setLevel(Level.ERROR);
22         //LOGGER.setLevel(Level.FATAL);
23         //LOGGER.setLevel(Level.INFO);
24         //LOGGER.setLevel(Level.TRACE);
25         //LOGGER.setLevel(Level.WARN);
26
27         double wordlength = 0.0;
28         double cnt = 0.0;
29
30         for (IntWritable value : values) {
31             wordlength += Double.valueOf(value.get());
32             cnt += 1.0;
33
34             LOGGER.warn(key + " --(Acc.)--> " + wordlength);
35         }
36         if(cnt > 0.0) {
37             wordlength = wordlength / cnt;
38         }
39         context.write(key, new DoubleWritable(wordlength));
40     }
41 }
42 }

```

AVOIDING EXPENSIVE OPERATIONS

Put a logger in the Reducer for outputting information.



CHECK THE LOG CONTENT


```
training@quickstart:~/workspace/logging/src
File Edit View Search Terminal Help
e.tar.gz lgwc_res
Programmer: Pei-Wei Tsai
Student ID: 92144979
24/01/13 05:17:35 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/01/13 05:17:35 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
24/01/13 05:17:35 INFO input.FileInputFormat: Total input paths to process : 1
24/01/13 05:17:36 INFO mapreduce.JobSubmitter: number of splits:1
24/01/13 05:17:36 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1704950385551_0013
24/01/13 05:17:36 INFO impl.YarnClientImpl: Submitted application application_1704950385551_0013
24/01/13 05:17:36 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1704950385551_0013/
24/01/13 05:17:36 INFO mapreduce.Job: Running job: job_1704950385551_0013
24/01/13 05:17:42 INFO mapreduce.Job: Job job_1704950385551_0013 running in uber mode : false
24/01/13 05:17:42 INFO mapreduce.Job:  map 0% reduce 0%
24/01/13 05:17:58 INFO mapreduce.Job:  map 100% reduce 0%
24/01/13 05:18:06 INFO mapreduce.Job:  map 100% reduce 100%
24/01/13 05:18:08 INFO mapreduce.Job: Job job_1704950385551_0013 completed successfully
```

Find Your
Job ID

5.13_with_code [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Sat Jan 13, 5:31 AM training

Applications Places System

JobHistory - Mozilla Firefox

JobHistory

quickstart.cloudera:19888/jobhistory/app

Hue HDFS

 JobHistory

Logged in as: dr.who

Retired Jobs

Show 20 entries

Search:

Submit Time	Start Time	Finish Time	Job ID	Name	User	Queue	State	Maps Total	Maps Completed	Reduces Total	Reduces Completed
2024.01.13 05:17:36 PST	2024.01.13 05:17:41 PST	2024.01.13 05:18:05 PST	job_1704950385551_0013	Logging Word Count	training	root.training	SUCCEEDED	1	1	1	1
2024.01.13 04:28:34 PST	2024.01.13 04:28:42 PST	2024.01.13 04:29:03 PST	job_1704950385551_0012	Word Count	training	root.training	SUCCEEDED	1	1	1	1
2024.01.13 03:50:07 PST	2024.01.13 03:50:21 PST	2024.01.13 03:50:42 PST	job_1704950385551_0008	Combiner Word Count	training	root.training	SUCCEEDED	1	1	1	1
2024.01.13 02:51:31 PST	2024.01.13 02:51:47 PST	2024.01.13 02:52:16 PST	job_1704950385551_0007	Word Count	training	root.training	SUCCEEDED	1	1	1	1
2024.01.12 23:36:37 PST	2024.01.12 23:36:47 PST	2024.01.12 23:37:12 PST	job_1704950385551_0006	Word Count	training	root.training	SUCCEEDED	1	1	1	1
2024.01.12 22:10:08 PST	2024.01.12 22:10:14 PST	2024.01.12 22:10:35 PST	job_1704950385551_0005	Average Word Length	training	root.training	SUCCEEDED	1	1	1	1
2024.01.12 21:55:19 PST	2024.01.12 21:55:37 PST	2024.01.12 21:56:16 PST	job_1704950385551_0004	Average Word Length	training	root.training	FAILED	0	0	0	0
2024.01.10 21:42:11 PST	2024.01.10 21:42:37 PST	2024.01.10 21:43:02 PST	job_1704950385551_0001	Word Count	training	root.training	SUCCEEDED	1	1	1	1

Showing 1 to 8 of 8 entries

First Previous 1 Next Last

training@quickstart: ~... java - logging/src/stub... JobHistory - Mozilla Fir...

LOGFILE
VIA WEB
UI



JobHistory

Logged in as: drwho

- Application
- About Jobs
- Tools

Retired Jobs

Show 20 entries												
Search:												
Submit Time	Start Time	Finish Time	Job ID	Name	User	Queue	State	Maps Total	Maps Completed	Reduces Total	Reduces Completed	
2024.01.13 06:41:28 PST	2024.01.13 06:41:32 PST	2024.01.13 06:41:51 PST	job_1704950385551_0021	LoggingAvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:39:26 PST	2024.01.13 06:39:31 PST	2024.01.13 06:39:48 PST	job_1704950385551_0020	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:35:52 PST	2024.01.13 06:35:55 PST	2024.01.13 06:36:28 PST	job_1704950385551_0019	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:33:46 PST	2024.01.13 06:33:49 PST	2024.01.13 06:34:06 PST	job_1704950385551_0018	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:31:56 PST	2024.01.13 06:32:00 PST	2024.01.13 06:32:20 PST	job_1704950385551_0017	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:28:57 PST	2024.01.13 06:29:05 PST	2024.01.13 06:29:31 PST	job_1704950385551_0016	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 06:05:07 PST	2024.01.13 06:05:16 PST	2024.01.13 06:06:09 PST	job_1704950385551_0015	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 05:48:18 PST	2024.01.13 05:48:25 PST	2024.01.13 05:48:48 PST	job_1704950385551_0014	AvgWordLength	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 05:17:36 PST	2024.01.13 05:17:41 PST	2024.01.13 05:18:05 PST	job_1704950385551_0013	Logging Word Count	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 04:28:34 PST	2024.01.13 04:28:42 PST	2024.01.13 04:29:03 PST	job_1704950385551_0012	Word Count	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 03:50:07 PST	2024.01.13 03:50:21 PST	2024.01.13 03:50:42 PST	job_1704950385551_0008	Combiner Word Count	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.13 02:51:31 PST	2024.01.13 02:51:47 PST	2024.01.13 02:52:16 PST	job_1704950385551_0007	Word Count	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.12 23:36:37 PST	2024.01.12 23:36:47 PST	2024.01.12 23:37:12 PST	job_1704950385551_0006	Word Count	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.12 22:10:08 PST	2024.01.12 22:10:14 PST	2024.01.12 22:10:35 PST	job_1704950385551_0005	Average Word Length	training	root.training	SUCCEEDED	1	1	1	1	
2024.01.12 21:55:19 PST	2024.01.12 21:55:37 PST	2024.01.12 21:56:16 PST	job_1704950385551_0004	Average Word Length	training	root.training	FAILED	0	0	0	0	

Find the Job in Web UI

5.13_with_code [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Sat Jan 13, 6:48 AM training

Applications Places System

MapReduce Job job_1704950385551_0013 - Mozilla Firefox

MapReduce Job job_170... x

quickstart.cloudera:19888/jobhistory/job/job_1704950385551_0013

Hue HDFS

hadoop

MapReduce Job job_1704950385551_0013

Logged in as: drwho

Application

Job

Overview

Counters

Configuration

Map tasks

Reduce tasks

Tools

Job Overview

Job Name: Logging Word Count

User Name: training

Queue: root.training

State: SUCCEEDED

Uberized: false

Submitted: Sat Jan 13 05:17:36 PST 2024

Started: Sat Jan 13 05:17:41 PST 2024

Finished: Sat Jan 13 05:18:05 PST 2024

Elapsed: 24sec

Diagnostics:

Average Map Time: 13sec

Average Shuffle Time: 3sec

Average Merge Time: 0sec

Average Reduce Time: 1sec

ApplicationMaster

Attempt Number	Start Time	Node	Logs
1	Sat Jan 13 05:17:37 PST 2024	quickstart.cloudera:8042	logs

Task Type	Total	Complete
Map	1	1
Reduce	1	1

Attempt Type	Failed	Killed	Successful
Maps	0	0	1
Reduces	0	0	1

training@quickstart:~...

java - logging/src/stub...

MapReduce Job job_17...

Right Ctrl

Find the Job
in Web UI



Map Tasks for job_1704950385551_0013

Logged in as: dr.who

Show 20 entries

Search:

Task							
Name	State	Start Time	Finish Time	Elapsed Time	Start Time	Finish Time	Elapsed Time
task_1704950385551_0013_m_000000	SUCCEEDED	Sat Jan 13 05:17:43 -0800 2024	Sat Jan 13 05:17:57 -0800 2024	13sec	Sat Jan 13 05:17:43 -0800 2024	Sat Jan 13 05:17:57 -0800 2024	13sec
ID	State	Start Time	Finish Time	Elapsed Time	Start Time	Finish Time	Elapsed Time

Showing 1 to 1 of 1 entries

FirstPrevious1NextLast

Find the Job in Web UI

Find the Job in Web UI

- Click the “Logs” to view the log file.

Show 20 entries		Search:						
Attempt	State	Status	Node	Logs	Start Time	Finish Time	Elapsed Time	Note
attempt_1704950385551_0013_m_000000_0	SUCCEEDED	map	/default-rack/quickstart.cloudera:8042	logs	Sat Jan 13 05:17:43 -0800 2024	Sat Jan 13 05:17:57 -0800 2024	13sec	
Attempt	State	Status	Node	Logs	Start Time	Finish Time	Elapsed Time	Note
Showing 1 to 1 of 1 entries					First Previous 1 Next Last			

Find the Job in Web UI

- Mapper Log File

The screenshot displays the Hadoop Web UI in a Mozilla Firefox browser window. The address bar shows the URL: `http://quickstart.cloudera:19888/jobhistory/logs/quickstart.cloudera:39797/container_1704950385551_0022_01_000002/attempt_1704950385551_0022_m_000000_0/training/syslog/`. The page title is "Hue HDFS". The left sidebar contains a navigation menu with "Application", "About", "Jobs", and "Tools". The main content area shows the job log for "training" on "Sat Jan 13, 6:55 AM". The log type is "syslog" and the log length is "62366633". The log content is a series of log entries from the Hadoop cluster, including warnings about configuration, information about the metrics system, and details about the mapreduce job execution. The log entries are formatted as `timestamp hostname:pid [main] org.apache.hadoop.metrics2.impl.MetricsConfig: Cannot locate configuration: tried hadoop-metrics2-maptask.properties,hadoop-metrics2.properties`. The log ends with `2024-01-13 06:53:59,534 INFO [main] stubs.LetterMapper: g --> 11`. The bottom of the screen shows the taskbar with the terminal window "training@quickstart:~...", the Java logging window "java - logging/src/stub...", and the Mozilla Firefox window.

5.13_with_code [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Sat Jan 13, 6:55 AM training

Applications Places System

Mozilla Firefox

http://quicks...log/?start=0

quickstart.cloudera:19888/jobhistory/logs/quickstart.cloudera:39797/container_1704950385551_0022_01_000002/attempt_1704950385551_0022_m_000000_0/training/syslog/

Hue HDFS

Logged in as: dr:who

hadoop

Application

About

Jobs

Tools

Log Type: syslog

Log Upload Time: Sat Jan 13 06:54:24 -0800 2024

Log Length: 62366633

2024-01-13 06:53:58,371 WARN [main] org.apache.hadoop.metrics2.impl.MetricsConfig: Cannot locate configuration: tried hadoop-metrics2-maptask.properties,hadoop-metrics2.properties

2024-01-13 06:53:58,443 INFO [main] org.apache.hadoop.metrics2.impl.MetricsSystemImpl: Scheduled snapshot period at 10 second(s).

2024-01-13 06:53:58,443 INFO [main] org.apache.hadoop.metrics2.impl.MetricsSystemImpl: MapTask metrics system started

2024-01-13 06:53:58,449 INFO [main] org.apache.hadoop.mapred.YarnChild: Executing with tokens:

2024-01-13 06:53:58,450 INFO [main] org.apache.hadoop.mapred.YarnChild: Kind: mapreduce.job, Service: job_1704950385551_0022, Ident: (org.apache.hadoop.mapreduce.security.token.JobTokenIdentifier@1e1a0406)

2024-01-13 06:53:58,586 INFO [main] org.apache.hadoop.mapred.YarnChild: Sleeping for 0ms before retrying again. Got null now.

2024-01-13 06:53:58,720 INFO [main] org.apache.hadoop.mapred.YarnChild: mapreduce.cluster.local.dir for child: /var/lib/hadoop-yarn/cache/yarn/nm-local-dir/usercache/training/appcache/application_1704950385551_0022

2024-01-13 06:53:58,934 INFO [main] org.apache.hadoop.conf.Configuration.deprecation: session.id is deprecated. Instead, use dfs.metrics.session-id

2024-01-13 06:53:59,214 INFO [main] org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter: File Output Committer Algorithm version is 1

2024-01-13 06:53:59,214 INFO [main] org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false

2024-01-13 06:53:59,230 INFO [main] org.apache.hadoop.mapred.Task: Using ResourceCalculatorProcessTree : []

2024-01-13 06:53:59,380 INFO [main] org.apache.hadoop.mapred.MapTask: Processing split: hdfs://quickstart.cloudera:8020/user/training/shakespeare.tar.gz:0+2808057

2024-01-13 06:53:59,478 INFO [main] org.apache.hadoop.mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)

2024-01-13 06:53:59,478 INFO [main] org.apache.hadoop.mapred.MapTask: mapreduce.task.io.sort.mb: 100

2024-01-13 06:53:59,478 INFO [main] org.apache.hadoop.mapred.MapTask: soft limit at 83886080

2024-01-13 06:53:59,478 INFO [main] org.apache.hadoop.mapred.MapTask: bufstart = 0; bufvoid = 104857600

2024-01-13 06:53:59,478 INFO [main] org.apache.hadoop.mapred.MapTask: kvstart = 26214396; length = 6553600

2024-01-13 06:53:59,481 INFO [main] org.apache.hadoop.mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask\$MapOutputBuffer

2024-01-13 06:53:59,499 INFO [main] org.apache.hadoop.io.compress.zlib.ZlibFactory: Successfully loaded & initialized native-zlib library

2024-01-13 06:53:59,500 INFO [main] org.apache.hadoop.io.compress.CodecPool: Got brand-new decompressor [.gz]

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: s --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 1 --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 5 --> 1

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: u --> 5

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 13

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: g --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: s --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: c --> 8

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 1 --> 11

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 6

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 1

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: u --> 5

2024-01-13 06:53:59,533 INFO [main] stubs.LetterMapper: 0 --> 13

2024-01-13 06:53:59,534 INFO [main] stubs.LetterMapper: g --> 11

training@quickstart:~... java - logging/src/stub... Mozilla Firefox

Right Ctrl

Find the Job in Web UI

- Reducer Log File

Log Type: stderr
Log Upload Time: Sat Jan 13 06:54:24 -0800 2024
Log Length: 0

Log Type: stdout
Log Upload Time: Sat Jan 13 06:54:24 -0800 2024
Log Length: 0

Log Type: syslog
Log Upload Time: Sat Jan 13 06:54:24 -0800 2024
Log Length: 78107366

Showing 4096 bytes of 78107366 total. Click [here](#) for the full log.

```
--> 76.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 80.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 86.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 93.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 95.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 101.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 105.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 109.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 116.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 120.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 127.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 134.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 138.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 142.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 146.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 150.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 154.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 161.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 165.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 169.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 173.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 177.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 183.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 187.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 191.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 197.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 201.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 205.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 208.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 212.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 216.0
2024-01-13 06:54:13,053 WARN [main] stubs.AverageReducer: z --(Acc.)--> 220.0
-----
```

Log Output Restrictions

If you suspect the input data of being faulty, you may be tempted to log the (key, value) pairs your Mapper receives.

- Reasonable for small amounts of input data.
- If your job runs across 500GB of input data, you could be writing up to 500GB of log files!
- Remember to think at scale when do so.

Instead of wrap vulnerable sections of code in *try{...}* blocks, write logs in the *catch {...}* blocks helps you only log down the critical data.



RETRIEVING JOB INFORMATION WITH COUNTERS

What are Counters?

Counters provide a way for Mappers or Reducers to pass aggregate values back to the driver after the job has completed.

- Their values are also visible from the JobTracker's Web UI.
- They are reported on the console when the job ends.

Very basic: just have a name and a value.

- Value can be incremented within the code.

What are Counters?

Counters are collected into Groups.

- Within the group, each Counter has a name.

Example: A group of counters called RecordType

- Names: TypeA, TypeB, TypeC
- Appropriate Counter can be incremented as each record is read in the Mapper.

Counters

Counters can be set and incremented via the method.

- `context.getCounter(group, name).increment(amount);`

Example

- `context.getCounter("RecordType","A").increment(1);`

What are Counters?

To retrieve Counters in the Driver code after the job is complete, use code such as examples below in the driver.

```
long typeARecords = job.getCounters().findCounter("RecordType","A").getValue();
```

```
long typeBRecords = job.getCounters().findCounter("RecordType","B").getValue();
```

Important!

Do NOT rely on a counter's value from the Web UI while a job is running.

- Due to possible speculative execution, a counter's value could appear larger than the actual final value.
- Modifications to counters from subsequently killed/failed tasks will be removed from the final count.



REUSE OF OBJECTS

Reuse of Objects

- It is generally good practice to reuse objects instead of creating many new objects.
- Creating new objects increases the usage of RAM space and overhead of memory allocation.

```
context.write(new Text(word.substring(0, 1)), new DoubleWritable(word.length()));
```

```

public class LetterMapper extends Mapper<LongWritable, Text, Text, DoubleWritable> {

    private static final Logger LOGGER = LogManager.getLogger(LetterMapper.class.getName());

    private final static DoubleWritable mapperValue = new DoubleWritable();
    private Text mapperKey = new Text();

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String tmp = context.getConfiguration().get("caseSensitive");
        Boolean caseSensitivie = (tmp.equalsIgnoreCase("true")? true : false);

        String line = (caseSensitivie? value.toString() : value.toString().toLowerCase());
    }
}

```

The Alternative

- A better practice is to create those frequently used but the value doesn't need to be kept for long objects outside of the method.


```
context.write(new Text(word.substring(0, 1)), new DoubleWritable(word.length()));
```



```
mapperValue.set(word.length());  
mapperKey.set(word.substring(0, 1));  
context.write(mapperKey, mapperValue);
```

The Alternative

- A better practice is to create those frequently used but the value doesn't need to be kept for long objects outside of the method.

```

public class AverageReducer extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    private static final Logger LOGGER = LogManager.getLogger(AverageReducer.class.getName());

    private final static DoubleWritable reducerValue = new DoubleWritable();

    @Override
    public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
        throws IOException, InterruptedException {

        double wordCount = 0;
        double cnt = 0;
        //LOGGER.setLevel(Level.WARN);

        for (DoubleWritable value : values) {

            wordCount += value.get();
            cnt += 1.0;

            LOGGER.warn(key + " --(Acc.)--> " + wordCount);
        }
        reducerValue.set((wordCount/cnt));
        context.write(key, reducerValue);
    }
}

```

The Alternative

- A better practice is to create those frequently used but the value doesn't need to be kept for long objects outside of the method.



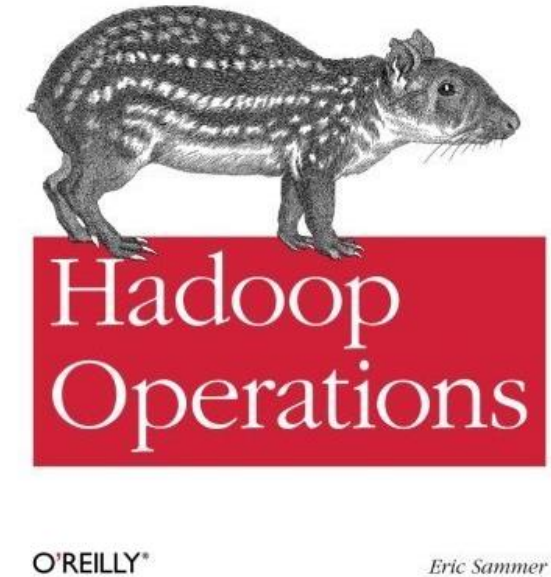
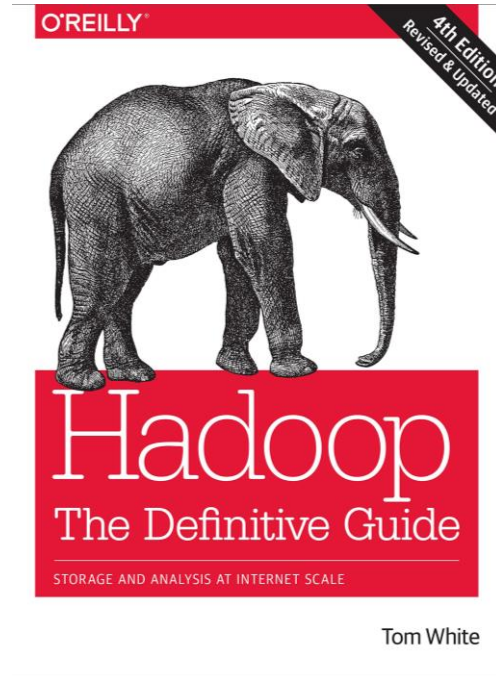
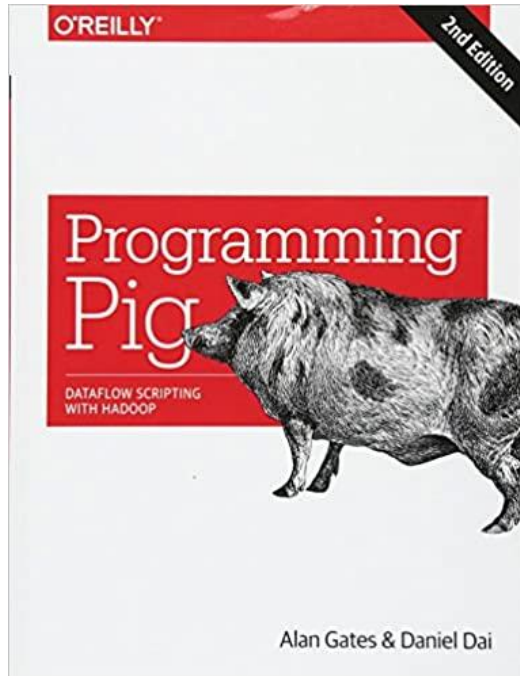
MAP-ONLY JOB

Map-Only MapReduce Jobs

- Many type of jobs, in practical, actually fit in the case that only Mapper is needed.
- Examples
 - File format conversion
 - Input data sampling
 - Image processing
 - Extract, Transform, Load (ETL) data processing methods

Map-Only MapReduce Jobs

- To create a Map-only job, set the number of Reducers to 0 in the Driver code by *job.setNumReduceTasks(0);*
- Call the *job.setOutputKeyClass* and *job.setOutputValueClass* methods to specify the output types.
- Anything written using the *context.write* method in the Mapper will be written to HDFS.
 - The original output should be written in the intermediate data.
 - One file per Mapper will be written.



Texts and Resources

- Unless stated otherwise, the materials presented in this lecture are taken from:
 - Hadoop: the definitive guide, White, Tom (Tom E.) author., 4th ed., Sebastopol, California: O'Reilly, 2015.
 - Hadoop operations, Sammer, Eric., Loukides, Michael Kosta.; Nash, Courtney.; Romano, Robert (Illustrator), illustrator., First edition., Sebastopol, CA: O'Reilly, 2012.
 - Programming pig: dataflow scripting with hadoop, Gates, Alan, author.; Dai, Daniel, 2nd edition., Beijing, China: O'Reilly, 2017



Unit Summary

- MapReduce Code Explanation.

Summary

